SPE-228892-MS

Optimizing Geosteering in Thin, Stacked Carbonate Clinoforms: A Step-Out Approach Using Advanced Tools to Reduce Uncertainty and Completion Risk

A. N. Al Hashami, M. M. Al Rashdi, M. S. Al Badi, S. E. Huisman, M. S. Al Hinai, I. H. Al Salmi, S. S. Al Rahbi, H. S. Maqbali, and A. Al Harthi, Petroleum Development Oman, Muscat, Sultanate of Oman

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Thin, stacked carbonate clinoform reservoirs within argillaceous formations present significant challenges for hydrocarbon development due to their complex depositional architecture, lateral discontinuities, and structural variability. Precise well placement is critical to maximize reservoir contact, avoid costly sidetracks, and reduce completion risk. This study presents an optimized geosteering methodology combining advanced real-time formation evaluation tools with a step-out well placement strategy aimed at reducing geological uncertainty and enhancing operational efficiency in such complex settings.

The development campaign involved drilling horizontal wells adjacent to existing producers to capitalize on reduced structural uncertainty and improve reservoir connectivity. A novel deep azimuthal resistivity tool, complemented by azimuthal electromagnetic sensors, provided high-resolution, real-time data that enhanced formation characterization and allowed dynamic adjustments to the well trajectory based on gamma ray, resistivity, and dip responses. Initial data from vertical and pilot wells informed the sweet zone criteria, which were iteratively refined as field experience accrued, particularly addressing challenges related to thin reservoir thickness (1-3 meters) and local dip variations.

Early wells encountered unexpected complexity, including multiple unintentional reservoir exits and increased sidetracking due to underestimated dip variability. However, the integration of advanced geosteering technology enabled more accurate structural interpretation and significantly improved in-zone well placement in later wells. These improvements translated into higher reservoir exposure, fewer sidetracks, reduced isolation completions, and sustained production rates, validating the geological model and guiding a structured field development plan involving optimized producer-injector well pairs.

This study offers a practical, replicable framework for geosteering in thin, laterally discontinuous carbonate clinoforms, demonstrating that even in greenfield environments with limited seismic and petrophysical data, precision geosteering coupled with adaptive operational workflows can overcome geological complexities. The findings underscore the importance of real-time decision-making, iterative learning, and the integration of advanced downhole tools to maximize recovery and reduce development risk in challenging carbonate reservoirs.

Introduction — Navigating Complexity in Carbonate Clinoforms

Carbonate clinoform reservoirs represent both high-value and high-risk targets in hydrocarbon exploration and development. Their complex depositional architectures—typically comprising thin, laterally discontinuous, and stacked beds—pose significant challenges for precise well placement. These challenges are amplified in argillaceous settings, where limited seismic resolution and structural ambiguity increase geological uncertainty and operational risk.

To address these challenges, geosteering has emerged as a critical enabler, allowing for real-time wellbore trajectory adjustments based on downhole formation evaluation data. However, in thin carbonate clinoforms, geosteering remains inherently difficult. Sweet zones are often narrower than predicted, and subtle dip variations can result in unintentional exits, leading to costly sidetracks or the need for isolation completions—both of which reduce overall development efficiency.

This paper presents a refined geosteering methodology tailored to the Upper Shuaiba Clinoforms in northern Oman. These clinoforms are characterized by thin, stacked carbonate reservoirs interbedded within argillaceous intervals, where conventional steering strategies often fall short. By integrating advanced, high-resolution geosteering tools with a step-out well placement strategy, the proposed approach systematically reduces structural and stratigraphic uncertainty. The result is a more reliable, iterative, and data-informed steering workflow that improves reservoir exposure, reduces sidetrack risk, and enhances drilling performance in complex geological settings.

The following sections describe the geological context, technical workflow, and field implementation of this methodology, culminating in a discussion of results, operational learnings, and broader implications for clinoform reservoir development

Geological Setting — Characterizing the Thin Carbonate Clinoforms

The Upper Shuaiba Clinoforms in northern Oman present a geologically complex and technically challenging reservoir setting. These clinoform systems are defined by progradational carbonate sedimentary structures that exhibit significant lateral facies variability and depositional heterogeneity. The presence of a thick, overlying Nahr Umr shale further complicates reservoir imaging and well placement due to its poor seismic resolution and mechanical contrast.

Stratigraphic analysis reveals that the Upper Shuaiba is predominantly composed of thin carbonate clinoform layers, typically ranging from 1 to 6 meters in thickness, interbedded with argillaceous strata. These deposits were formed during a regressive phase in the late Aptian, when falling sea levels led to the development of clinoform-fringing shoals around the Bab Basin (Figure 1). These shoals advanced basinward in concentric arcs toward the southeast, south, and southwest, forming a series of stacked carbonate wedges with complex internal architecture and varying sediment composition—from clean carbonates to clay-rich facies.

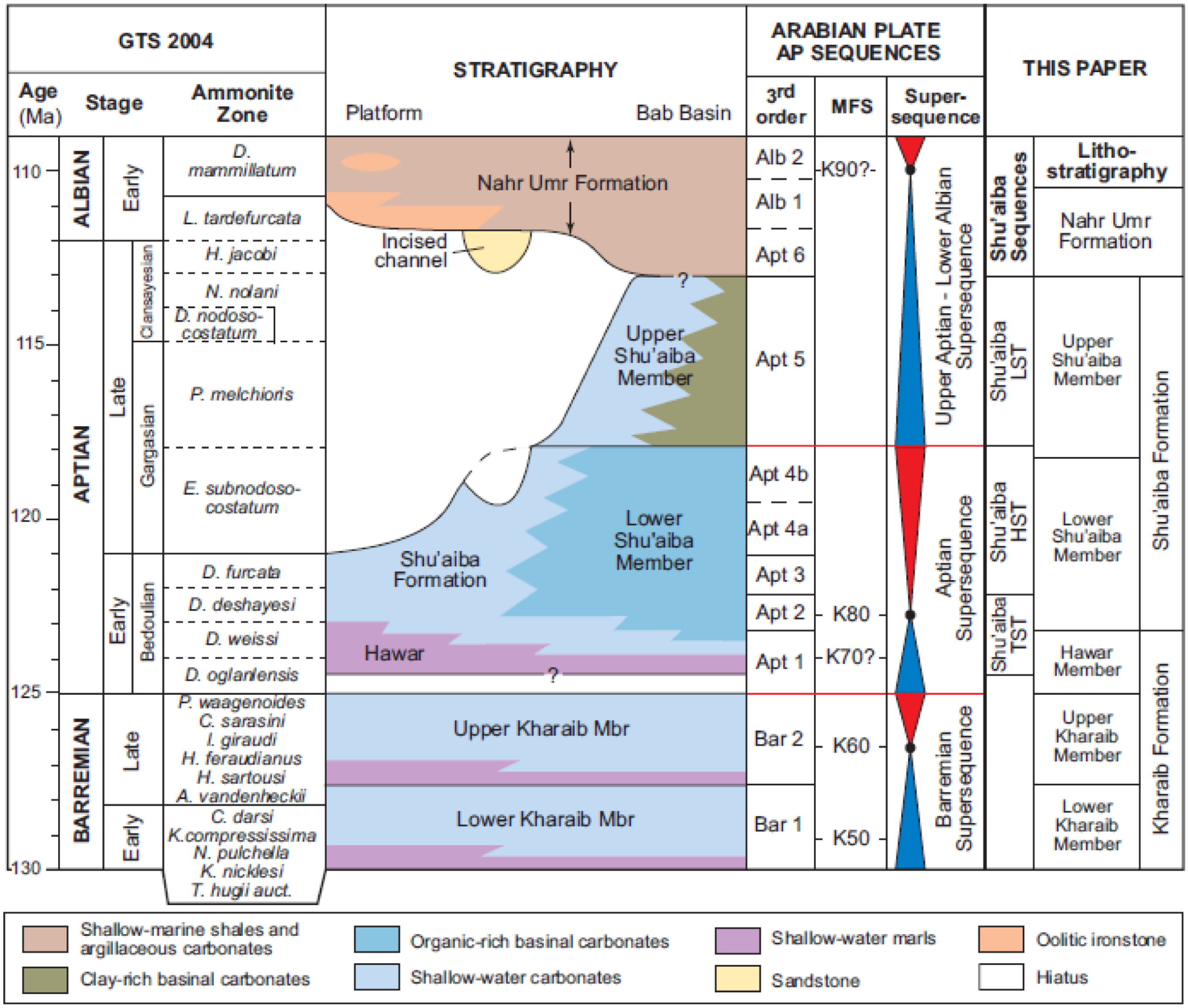

Figure 1—Chronostratigraphic reference column for the eastern Arabian Plate (van Buchem et al., 2010) with lithostratigraphy for the Barremian to Albian in the Sultanate of Oman (Henk J. Droste., 2010).

Within the clinoform system, multiple progradational units can be distinguished, each characterized by different rock types and reservoir qualities. A widely adopted core-based rock type classification divides these into five categories:

- RT-A: Grainstones and grain-rich packstones
- RT-B: Rudist-bearing carbonates
- RT-C: Microporous ramp facies
- RT-D: Slightly clayey carbonates
- RT-E: Clay-dominated, low-permeability units

Reservoir quality is generally highest in RT-A, RT-B, and RT-C, which form the primary hydrocarbon-bearing intervals. In contrast, RT-D and RT-E commonly act as barriers or baffles due to their low permeability and poor hydrocarbon mobility.

A key reservoir unit within the formation is the Upper Shuaiba B (USB), which is further divided into two sub-zones—B1 and B2. The B1 zone typically offers better porosity and permeability, whereas B2 is denser and less productive. However, due to the thinness of B1, horizontal wells often traverse both sub-zones to ensure sufficient reservoir contact and optimize production.

Production data from the field suggests that the rudist-rich and microporous facies dominate the USB reservoir interval. The only vertical well drilled into this unit confirmed a predominantly rudist composition. Notably, observed low oil saturation is attributed more to well positioning within a lower-quality part of the reservoir than to inherent rock properties.

The geological complexity of the Upper Shuaiba—particularly the thin bedding, lateral facies changes, and stratigraphic pinch-outs— poses a considerable challenge to both reservoir characterization and horizontal well placement. Studies by Raghunathan et al. (2009) and more recently Al-Hamdi, Nicholls, and Azzazi (2023) underscore the importance of understanding lateral facies offset and local dip variation when forecasting productivity and defining steering strategies.

Figure 2 shows a representative NW-SE cross-section across the study area, illustrating the clinoform architecture and depositional evolution of the Upper Shuaiba within the Bab Basin. These complex structural and stratigraphic features significantly influence fluid flow behavior, reservoir continuity, and ultimately recovery efficiency.

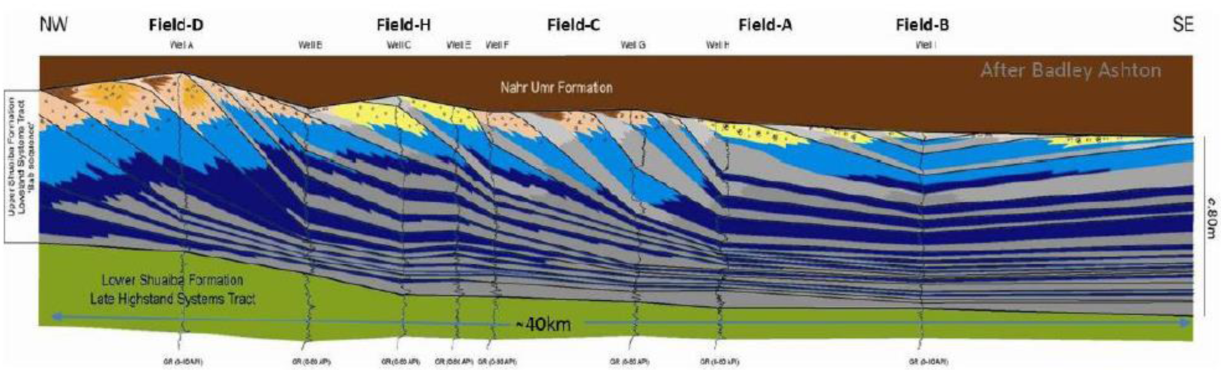

Figure 2—NW-SE geological cross and associated regional depositional model of the Upper Shuaiba clinoforms as part of the infill in the late Aptian-aged Bab Basin

In this context, the application of advanced geosteering technologies has become essential for improving wellbore placement and reservoir contact. Real-time interpretation of formation data has enhanced geological understanding, enabled more responsive trajectory control, and significantly improved development outcomes. The next section outlines the integrated geosteering methodology employed in this field to address these geological complexities.

Precision Geosteering in Complex Clinoforms — Tools and Execution

Optimizing well placement within the Upper Shuaiba Clinoforms presents a formidable technical challenge, primarily due to the extremely thin and laterally heterogeneous nature of the reservoir units. In the current Area of Interest (AoI), the primary target—the B1 unit—ranges from 1 to 3 meters in thickness, offering minimal vertical margin for error during horizontal drilling operations.

Development in this sector commenced with the drilling of a pilot well designed to evaluate reservoir quality and inform the placement of future laterals. The pilot results were promising porosity and saturation values within the B1 clinoform indicated a 2-meter-thick hydrocarbon-bearing interval. Based on this data, a sidetrack was initiated to drill a horizontal section targeting the B1 reservoir (Figure 2).

However, the lateral section underperformed relative to expectations. Despite multiple azimuthal corrections during drilling—aimed at replicating the favorable pilot results—log responses showed a significant reduction in both porosity and saturation. The initial interpretation concluded that the reservoir in this area was of poor quality and uneconomic for further development. The well was nevertheless completed and brought online.

Surprisingly, the well continued producing well beyond forecasted levels, maintaining stable output over a three-year period. This unexpected performance prompted a comprehensive re-evaluation of the reservoir and well placement strategy. A post-drill review suggested that the lateral section had likely missed the reservoir's central core, and that the poor log response was attributable to nonoptimal placement, rather than true reservoir quality.

This realization catalyzed a revised approach: a second well was proposed, employing advanced geosteering technologies to ensure more accurate in-zone placement. The objective was to maintain the trajectory within the most productive portion of the B1 unit by leveraging real-time formation evaluation data, thereby maximizing hydrocarbon exposure and minimizing the risk of sweet zone exit.

To enhance geosteering precision, deep azimuthal resistivity tools were selected to distinguish the B1 reservoir from the overlying Nahr Umr shale, using resistivity contrast as a steering discriminator. The tables (Table 1 & Table 2) below detail the investigation depth of both the conventional and new tools and resistivity values of the formations. These tools offer deep-looking capabilities that allow for early detection of bed boundaries, enabling proactive trajectory corrections. In parallel, azimuthal electromagnetic (EM) tools were deployed to map lateral facies transitions, detect structural heterogeneities (e.g., truncations, minor faults), and assess bedding orientation—critical information for navigating within thin, truncated clinoform packages.

Table 1—Depth of investigation for the tools.

Tool	Depth of investigation
Standard	5.5m (18 ft)
Advanced	9.1m (30fr)

Table 2—Resistivity values for the formation.

Formation	Resistivity values
Nahr Umr	~1 Ω.m
US B1	5-10 Ω.m

Together, these tools provided real-time geological feedback, allowing the well path to be dynamically adjusted based on evolving subsurface conditions. This approach significantly improved reservoir contact and minimized the risk of drilling out of zone, which is a common failure mode in such complex systems.

Just as critical to success was the integration of operational learnings from the initial well. This included:

- Refining the geological model based on production history and updated structural interpretations.
- Reassessing correlations between pilot well logs and lateral responses.
- Implementing improved azimuth control logic and steering algorithms.

Additionally, the team prioritized continuous performance monitoring and adaptive decision-making throughout the drilling process, ensuring that new information was incorporated in near real-time.

After successfully deploying the geosteering tools and achieving precise placement in the reservoir, two additional wells were drilled using the same methodology (Figure 3). These wells confirmed the robustness of the approach, delivering consistent results in terms of accurate geosteering and favorable reservoir properties. This reinforced confidence in the workflow and demonstrated its repeatability across the field.

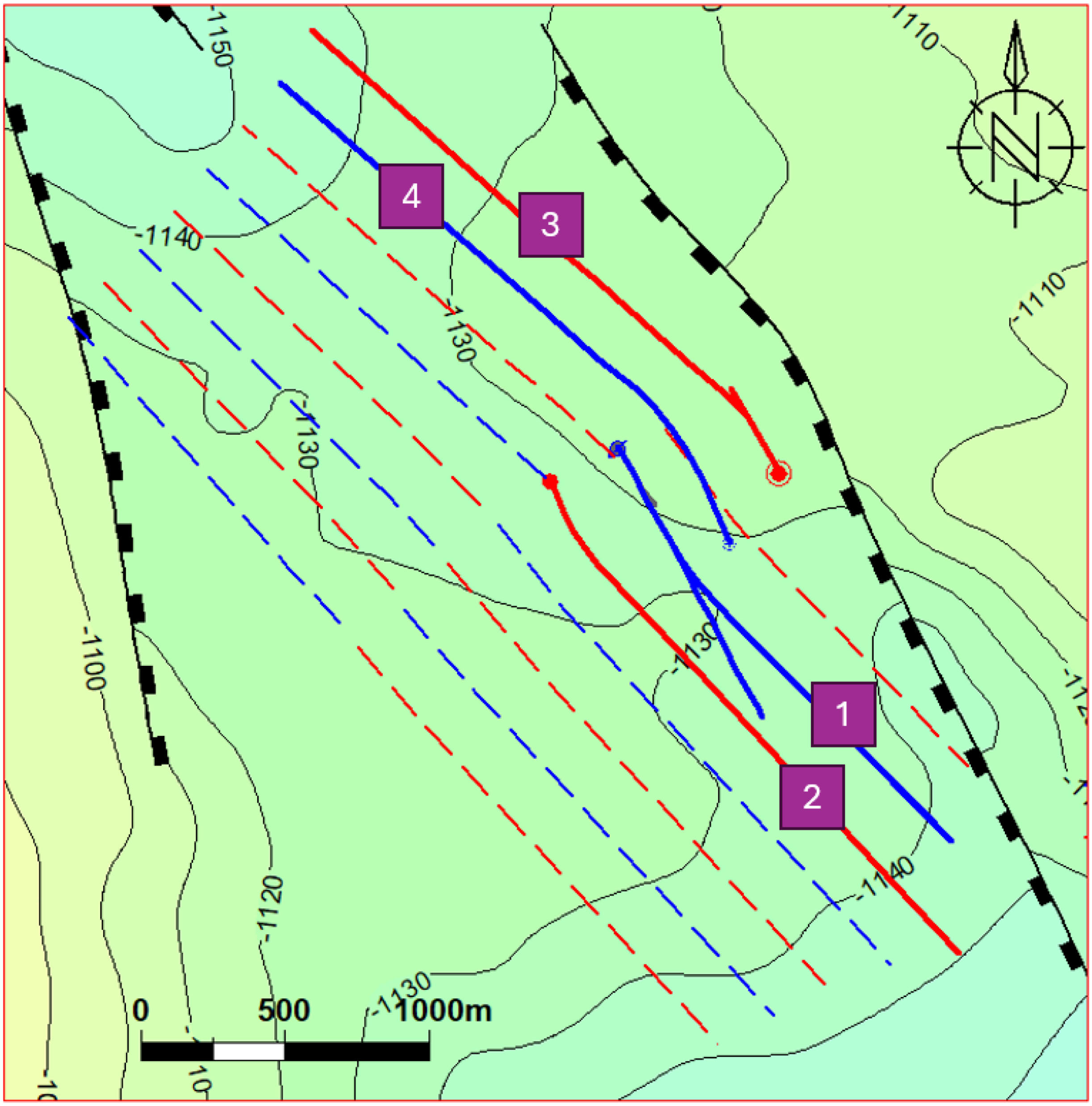

Figure 3—The structure map displays the locations of all drilled wells. The first well was drilled using the standard geosteering approach, while the second, third, and fourth wells employed the new geosteering tool and methodology. Following the successful drilling of these wells, this area of the field was de-risked and confirmed for development, enabling the planned drilling of additional wells.

This case underscores the importance of high-resolution formation evaluation, adaptive steering workflows, and data-driven decisionmaking in overcoming the inherent uncertainties of drilling in thin, laterally variable carbonate reservoirs. It also illustrates how postdrill production data—when systematically analyzed—can challenge initial assumptions and unlock overlooked reservoir potential.

In summary, the experience in this portion of the Upper Shuaiba Clinoforms highlights a clear evolution in well placement strategies— from conventional, pilot-based planning to a responsive, tool-enabled, and iterative geosteering methodology. This progression not only enhances placement accuracy but also supports long-term production optimization in structurally and stratigraphically complex reservoirs.

Results and Operational Insights from Field Implementation

The successful implementation of advanced geosteering technologies in the Upper Shuaiba Clinoforms is evident through the comparative performance of two offset wells, each drilled using different steering strategies. The results clearly demonstrate the operational and economic advantages of precision well placement in thin, laterally variable carbonate reservoirs.

Lessons from Initial Well Placement

A detailed post-drill analysis of the first horizontal well revealed suboptimal reservoir positioning. Although the well did intersect the B1 clinoform, its trajectory veered toward the upper margin of the unit, thereby missing the higher-quality central zone. This misplacement stemmed primarily from limited structural control and inadequate pre-drill dip data, which increased the risk of drilling too close to the overlying Nahr Umr shale—a potential seal breach risk.

In response, the drilling team revised the strategy for subsequent wells, aiming for a trajectory more centrally aligned within the B1 clinoform. This adjustment leveraged advanced real-time geosteering tools to reduce geological uncertainty and enhance reservoir exposure.

Deployment of Deep Resistivity and Azimuthal EM Tools

The follow up wells incorporated a combination of deep azimuthal resistivity and azimuthal electromagnetic (EM) sensors, as detailed in Al Araimi et al. (2024), which significantly improved trajectory control and subsurface visualization. Example of a cross-section illustrating the well drilled using advanced tools (Figure 4):

- The deep resistivity tool exploited the distinct contrast between the low-resistivity Nahr Umr Formation (~1 Ω·m) and the higher-resistivity B1 reservoir (5-10 Ω·m). This enabled real-time detection of both the reservoir's top and base, supporting proactive steering.
- The azimuthal EM sensors extended the depth of investigation beyond the immediate borehole, allowing the well to respond dynamically to dip steepening, lateral facies transitions, and structural truncations.

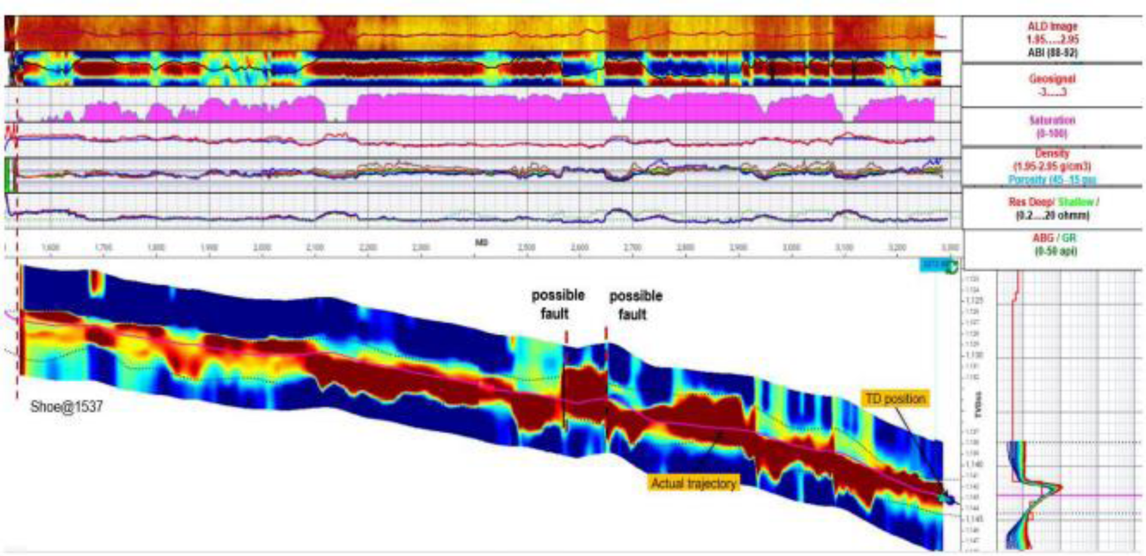

Figure 4—Leveraging advanced geosteering technology offered critical insights into formation dip, thickness, and proximity to contrasting boundaries, thereby enhancing well placement accuracy, improving outcomes, and enabling extended total depth (TD).

These tools proved especially valuable in detecting the abrupt geometries and small-scale dip variations typical of the Upper Shuaiba Clinoforms—features that conventional LWD measurements often miss.

Performance Comparison and Key Findings

The second well maintained a stable in-zone trajectory from landing point (LP) to total depth (TD), despite encountering a true vertical depth (TVD) variation exceeding 10 meters over a 1,500-meter lateral section (Figure 3). Crucially, this trajectory was achieved without any sidetracks.

In contrast, a nearby offset well drilled with conventional geosteering experienced multiple unanticipated structural deviations, exited the target zone several times, and ultimately required three sidetracks to re-enter the reservoir. This not only increased drilling time and cost, but also compromised wellbore integrity and completion effectiveness.

Figure 3. Real-time azimuthal resistivity data from the second well, showing continuous monitoring of formation dip, reservoir boundaries, and thickness, enabling improved wellbore placement and steering accuracy.

Development Strategy and Field Expansion

Encouraged by the success of the second well, the operator launched a full-field development plan for the targeted area. The strategy involves drilling over ten producer-injector pairs, spaced approximately 200 meters apart, covering a lateral extent of ~1.5 kilometers (Figure 4).

The structured development approach is designed to:

- Maximize reservoir drainage through optimal well spacing
- Ensure uniform pressure support via aligned injector-producer pairs
- Maintain trajectory control within the high-quality B1 flow unit
- Minimize sidetracks and avoid isolation completions, thereby reducing operational cost and risk

Figure 4. Planned spatial layout of development wells showing producer-injector pairs, designed for enhanced sweep efficiency and field management.

Strategic Implications

The second well's performance provided validation of the updated geological and petrophysical models, and demonstrated the tangible value of integrated geosteering workflows. Key takeaways include:

- Increased reservoir exposure, resulting in improved initial production and delayed water breakthrough
- Reduced drilling risk, with fewer sidetracks and lower non-productive time
- Improved geological model calibration, which enhances the planning of future wells and supports better reservoir management

In summary, the application of deep resistivity and azimuthal EM tools translated directly into improved economic and operational outcomes. This case study reinforces the strategic importance of precision geosteering in unlocking the development potential of thin, laterally discontinuous carbonate reservoirs, such as those found in the Upper Shuaiba Formation.

Discussion

The comparative performance of the two offset wells underscores the transformative impact of advanced geosteering technologies on well placement, operational efficiency, and reservoir management in geologically complex carbonate clinoforms.

The first well, drilled using conventional geosteering techniques, suffered from limited real-time structural resolution, resulting in a trajectory that intersected only the upper edge of the B1 reservoir unit. This suboptimal positioning reduced effective reservoir exposure and production potential, while increasing the risk of encroachment into the overlying Nahr Umr shale —a major concern in thin, dipping clinoform settings where vertical margins are narrow and structural dip is unpredictable.

In contrast, the second well—drilled using a fully integrated geosteering workflow—demonstrated the tangible value of deep azimuthal resistivity and azimuthal electromagnetic (EM) technologies. These tools enabled real-time identification of resistivity contrasts between the B1 reservoir and the bounding shale, as well as advanced dip detection and boundary mapping beyond the immediate borehole environment. This improved visibility was critical in navigating structural truncations and maintaining an optimal in-zone trajectory, even across a true vertical depth variation exceeding 10 meters.

Operational and Strategic Insights

This case highlights several key advantages associated with precision geosteering:

- Reduced Drilling Risk: Real-time formation tracking allowed the second well to maintain continuous reservoir contact without requiring sidetracks. This significantly reduced non-productive time (NPT), avoided wellbore integrity concerns, and enhanced operational safety.
- Economic Efficiency: By eliminating sidetracks and shortening the drilling timeline, the well achieved lower overall cost and faster time to production—delivering immediate benefits to project economics.
- Enhanced Reservoir Management: Optimal well placement enabled improved reservoir drainage and delayed water breakthrough, supporting longer plateau production periods and increased recovery efficiency.

Implications for Field Development Strategy

The field development plan—anchored on learnings from the second well—marks a strategic pivot toward data-driven, technology- enabled drilling. The proposed layout of injector-producer pairs is designed to ensure efficient pressure support and areal sweep, while real-time steering data will be used to continuously refine geological interpretations and update placement decisions on a well-by-well basis.

These results suggest that geosteering is no longer simply a drilling tool, but a strategic enabler of reservoir development in structurally and stratigraphically complex plays. The integration of real-time formation evaluation, steering logic, and operational feedback loops has become essential for unlocking tight or thinly bedded carbonate reserves with high lateral variability.

Limitations and Opportunities for Further Improvement

While advanced geosteering tools have significantly reduced placement uncertainty, they are not without limitations. Challenges remain in areas with ambiguous resistivity signatures, rapid lateral facies transitions, or unexpected faults not captured in pre-drill models.

To further improve performance, future development campaigns should emphasize:

- Continuous model calibration using downhole steering data and production history
- Enhanced integration of geological, petrophysical, and drilling data streams

- Exploration of additional technologies, such as real-time seismic while drilling (SWD) or high-resolution ultra-deep EM logging, to expand formation visibility and reduce subsurface ambiguity

In summary, the discussion emphasizes that precision geosteering is not merely a drilling tool but a critical component of reservoir development strategy, particularly in heterogeneous, structurally complex formations like the Upper Shuaiba clinoforms.

Conclusion

The successful application of advanced geosteering technologies in the Upper Shuaiba clinoforms highlights the critical importance of high-resolution formation evaluation in the effective development of complex, thin carbonate reservoirs. By integrating deep azimuthal resistivity and azimuthal electromagnetic (EM) tools within a real-time geosteering workflow, well trajectories were precisely maintained within the reservoir's most productive zones despite significant structural and stratigraphic variability.

The comparative analysis of offset wells revealed marked performance improvements with the advanced geosteering approach, including superior reservoir contact, elimination of costly sidetracks, and enhanced production outcomes. These results not only validated predrill geological models but also provided essential feedback to continuously refine future drilling strategies.

Furthermore, the evolution from isolated well successes to a comprehensive field development plan, centered around a structured producer-injector well pattern, demonstrates how geosteering innovations can drive long-term reservoir management and optimize hydrocarbon recovery efficiency.

Ultimately, this case study underscores the strategic value of precision geosteering in mitigating drilling risk, maximizing economic returns, and unlocking development potential in challenging carbonate settings. As reservoirs grow increasingly complex, the integration of advanced geosteering tools will remain a cornerstone of effective, sustainable hydrocarbon development.

Acknowledgments

I would like to extend my heartfelt appreciation to the entire project team for their dedication and collaborative spirit throughout the development of the Upper Shuaiba reservoir. A special note of thanks goes to our team leader for his steadfast support, guidance, and leadership, all of which have been instrumental to the success of this work.

We also gratefully acknowledge the Ministry of Energy and Minerals of the Sultanate of Oman for their continued support, and the management of Petroleum Development Oman LLC for granting the necessary permission to publish this paper and for their ongoing efforts in advancing the nation's energy sector.

In addition, we recognize the valuable contributions of our partner organizations, technical experts, and service providers, whose knowledge and innovative solutions played a vital role in the successful execution of the project. Finally, we thank the conference organizers and reviewers for their insightful feedback, which has helped strengthen the quality and clarity of this publication.

Nomenclature

EM – Electromagnetic
LWD – Logging While Drilling
TVD – True Vertical Depth
TD – Total Depth
LP – Landing Point
AoI – Area of Interest
USB – Upper Shuaiba B
RT-A to RT-E – Rock Type classification from grainstone to clay-rich facies

NPT – Non-Productive Time
GR – Gamma Ray
Hz – Hertz (frequency unit in EM logging)
Ω· m – Ohm-meter (unit of resistivity)
FID – Final Investment Decision
P10 / P50 / P90 – Probabilistic forecast percentiles used in production modeling
SPE – Society of Petroleum Engineers

References

Al Araimi, A., Al Yahyai, H., and Taher, A. (2024). Optimizing well placement in heterogeneous carbonate reservoir utilizing a new generation of high definition mapping tool. SPE Conference at Oman Petroleum & Energy Show, Muscat, Oman, 22-24 April 2024. SPE-218651-MS.

Al-Hamdi, A., Nicholls, C., and Azzazi, M., 2023, "Integrated workflow for understanding the sedimentary basin and the rock quality of the Cretaceous Upper Shuaiba, Bab Member, in the northwest of Oman through thickness mapping and rock typing at regional scale," paper SPE-216156-MS, in Proceedings of the Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, November 13-16, https://doi.org/10.2118/216156-MS.

Raghunathan, Murali, Yousuf Salim Al-Zaabi, Anouk Creusen, Mariano Carrera, Maniesh Singh, Richard Raymond Spiteri, Harry Soek, Roeland Christiaan Roeterdink, and Paula Wigley. 2009. Fast Appraisal And Development Of A Stratigraphic Carbonate Play. Paper presented at the International Petroleum Technology Conference, Doha, Qatar: doi: https://doi.org/10.2523/IPTC-13883-MS.